

Erik Nordquist

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20 Penn St, Baltimore, MD 21201

Education

- 2023 **Ph.D.** Chemistry, University of Massachusetts Amherst.
2018 **B.S.** Chemistry and Physics, The College of Idaho, *summa cum laude*.

Appointments

- 2023– **Postdoctoral Fellow**, University of Maryland, Baltimore, Department of Pharmaceutical Sciences.
2025– **Adjunct Professor**, Stevenson University, Department of Chemistry.
Collaborative Teaching Fellowship, University of Maryland–Stevenson partnership.

Fellowships

- 2023–26 **NIH T32 Postdoctoral Fellowship**, University of Maryland, Baltimore.
2022 **Teaching Fellowship**, College of Natural Sciences, University of Massachusetts Amherst.
Competitive fellowship supporting the design and teaching of a First-Year Seminar course.
2020–22 **NIH T32 Predoctoral Fellowship**, University of Massachusetts Amherst.

Research Experience

- 2023– **Postdoctoral Fellow**, University of Maryland Baltimore, Department of Pharmaceutical Sciences.
- **Advisor:** Alexander D. MacKerell, Jr.
 - Method development and collaborations in computer-aided drug design (CADD), including: PROTACs, covalent inhibitors, protein binding site identification, pharmacophore modeling.
- 2018–23 **Graduate Research Assistant**, University of Massachusetts, Amherst, Department of Chemistry.
- **Advisor:** Jianhan Chen; **Thesis:** Understanding Protein Structure-Function Relationships Using Simulations and Machine Learning
 - BK channel hydrophobic dewetting mechanism, free energy simulation method development.
 - Predictive models of protein function using molecular simulations and machine learning.
- 2017 **Undergraduate Researcher**, NSF REU, Boise State University, Department of Materials Science.
- **Advisor:** Rick Uvic. **Project:** Fabrication of hybrid organic-inorganic perovskite solar cells.
 - Instrumentation used: scanning electron microscope, x-ray diffractometer.
- 2014–16 **Summer Research Technician**, Northwest Irrigation and Soils Research Lab, USDA. Kimberly, ID.
- **Advisor:** Gary Lehrs. **Projects:** Effect of long-term manure on soil stability, and water infiltration.
 - Aggregate stability and water infiltration instruments, and data analysis.

Teaching Experience

Stevenson University

- 2025– **Instructor of record**, General Chemistry Lab, Stevenson University. Full teaching responsibility for the lab class, including grading and office hours. Fall 2025, Spring 2026, Fall 2026 (ongoing).

University of Maryland, Baltimore

2023–26 **Guest lecture**, Computer-Aided Drug Design in a graduate Cancer Biology course (4x);

University of Massachusetts Amherst

- 2022 **Instructor of record**, First-Year Seminar, Title: “Reconciling Atomic Chaos and Human Order.” I designed this course through the College of Natural Science Teaching Fellowship, including a two-day teaching seminar. ([link](#)) **Course goals**: develop scientific communication skills and perform 4-week research project. Assessments were primarily formative with self-feedback to build growth mindsets. **Guest lecture**, Discussion seminar on AlphaFold2 in a Biophysics course at Amherst College.
- 2020 **Guest lecture**, Graduate Statistical Mechanics lecture on molecular mechanics and simulation.
- 2018–19 **Graduate Teaching Assistant**, General Chemistry lab. Responsibilities: in-class instruction, grading, office hours, and exam proctoring.

Mentoring Experience

University of Massachusetts

- Samantha Schultz, undergraduate (2020–2021); now PhD student at UPenn.
- Callie Jillson, undergraduate (2019–2020); now Registered Nurse in MA.

University of Maryland

- Ruchira Basu, grad. student (2026–)
- Dami Showunmi, undergraduate, (2025–26)
- Anthony O'Donnell, grad. student (2024–)

Publications

#: Contributed equally.

Undergraduate mentees underlined.

Peer-reviewed:

- **Nordquist E**, MacKerell A. Automated pharmacophore hypothesis generator for SILCS-Pharm: ph4gen. **J. Comput. Aided Mol. Des.** 2026. DOI: [10.1007/s10822-026-00957-5](https://doi.org/10.1007/s10822-026-00957-5)
- Chan A[#], Eberly C[#], Goodis C[#], **Nordquist E**, Weldemariam M, Cooper B, Pogash S, Lapidus R, Kane M, MacKerell A, Civin C, Fletcher S. Proteolysis Targeting Chimera (PROTAC) Linkerology Enhances the CDK-Degradation Selectivity Profile of a Multi-CDK Inhibitor: Discovery of Potent, Anti-Leukemic CDK9 Degradators. **ACS Pharmacol. Transl. Sci.** 2026. DOI: [10.1021/acsptsci.6c00232](https://doi.org/10.1021/acsptsci.6c00232)
- **Nordquist E**, Zhao M, Yu W, MacKerell A. Computational Modeling of PROTAC Ternary Complexes as Ensembles Using SILCS-xTAC. **J. Chem. Inf. Model.** 2025. DOI: [10.1021/acs.jcim.5c02045](https://doi.org/10.1021/acs.jcim.5c02045)
- Patel R, **Nordquist E**, Polli J. Prediction of surfactant-mediated dissolution of poorly soluble drugs from drug particles. **Eur. J. Pharm. Sci.** 2025. DOI: [10.1016/j.ejps.2025.107052](https://doi.org/10.1016/j.ejps.2025.107052)
- **Nordquist E**[#], Zhao M[#], Kumar A, MacKerell A. Physics- and Machine-Learning Based Method to Identify Druggable Binding Sites Using SILCS-Hotspots. **J. Chem. Inf. Model.** 2024. DOI: [10.1021/acs.jcim.4c01189](https://doi.org/10.1021/acs.jcim.4c01189)
- **Nordquist E**, Jia Z, Chen J. Small Molecule NS11021 Promotes BK Channel Activation by Increasing Inner Pore Hydration. **J. Chem. Inf. Model.** 2024. DOI: [10.1021/acs.jcim.4c01012](https://doi.org/10.1021/acs.jcim.4c01012)
- **Nordquist E**[#], Zhang G[#], Barethiya S, Ji N, White K, Han L, Jia Z, Shi J, Cui J, and Chen J. Incorporating physics to overcome data scarcity in predictive modeling of protein function: a case study of BK channels. **PLOS Comput. Biol.** 2023. DOI: [10.1371/journal.pcbi.1011460](https://doi.org/10.1371/journal.pcbi.1011460)
- Zhang L, Barethiya S, **Nordquist E**, Chen J. Machine Learning Generation of Dynamic Protein Conformational Ensembles. **Molecules** 2023. DOI: [10.3390/molecules28104047](https://doi.org/10.3390/molecules28104047)
- **Nordquist E**, Zhiguang J, Chen J. Inner pore hydration free energy controls activations of the big potassium channel and its mutants. **Biophys. J.** 2023. DOI: [10.1016/j.bpj.2023.02.005](https://doi.org/10.1016/j.bpj.2023.02.005)

- **Nordquist E[#]**, **Schultz S[#]**, and Chen J. Using Metadynamics To Explore the Free Energy of Dewetting in Biologically Relevant Nanopores. **J. Phys. Chem. B** 2022 DOI: [10.1021/acs.jpcb.2c04157](https://doi.org/10.1021/acs.jpcb.2c04157)
- **Nordquist E**, Clerico E, Chen J, Gierasch L. Computational Modeling of Hsp70-Client Interactions: Past, Present, and Future. **J. Phys. Chem. B** 2022. DOI: [10.1021/acs.jpcb.2c03806](https://doi.org/10.1021/acs.jpcb.2c03806)
- **Nordquist E**, English C, Clerico E, Sherman W, Gierasch L, Chen J. Physics-based modeling provides predictive understanding of selectively promiscuous substrate binding by Hsp70 chaperones. **PLOS Comput. Biol.** 2021. DOI: [10.1371/journal.pcbi.1009567](https://doi.org/10.1371/journal.pcbi.1009567)
- Gong X, Chiricotto M, Liu X, **Nordquist E**, Feig M, Brooks C, Chen J. Accelerating the generalized born with molecular volume and solvent accessible surface area implicit solvent model using graphics processing units. **J. Comput. Chem.** 2020. DOI: [10.1002/jcc.26133](https://doi.org/10.1002/jcc.26133)

Book Chapters:

- **Nordquist E**, Horrigan F, MacKerell A. Computational ligand binding site prediction. **Springer Nature**. 2026. DOI: [10.1007/978-3-032-07511-6_7](https://doi.org/10.1007/978-3-032-07511-6_7)

Presentations

Conference Presentations

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| 2025 | Talk , Frontiers at the Chemistry-Biology Interface Symposium, Johns Hopkins University, "Computational modeling of ternary complexes as ensembles using SILCS-PROTAC," Baltimore, MD. |
| 2024 | Talk , Institute for Bioscience and Biotechnology Research, University of Maryland Baltimore, Early-Career Research Symposium, "Computational design of PROTACs", Rockville, MD.
Talk , University of Maryland Greenbaum Comprehensive Cancer Center Research Symposium, "Physics- and machine-learning-based method for identifying druggable binding sites with SILCS-Hotspots." (Talk award) Baltimore, MD.
Poster , Biophysical Society Annual Meeting, "Computational mapping of allosteric modulators of the BK channel." Philadelphia, PA. |
| 2023 | Poster , Biophysical Society Annual Meeting, "A predictive model of voltage gating of BK channels via physical modeling and machine learning." San Diego, CA. |
| 2022 | Poster , Biophysical Society Annual Meeting, "Free energy of hydrophobic dewetting in gating of BK channels." San Francisco, CA.
Talk , University of Massachusetts Amherst ResearchFest: "Predicting protein function with physics, experiments and machine learning." Amherst, MA. |
| 2020 | Talk , Northeastern Structural Symposium, "Physical origins of selective promiscuity to Hsp70s revealed through physics-based modeling." Virtual.
Poster , University of Massachusetts Amherst ResearchFest (WE McEwen Award): "Physical origins of selective promiscuity to Hsp70s revealed through physics-based modeling." Amherst, MA. |
| 2019 | Talk , Biophysics at University of Massachusetts Amherst, "Understanding the origins of DnaK's selective promiscuity with physics-based modeling." Amherst, MA.
Poster , Molecular Biophysics in the Northeast, "Understanding the origins of DnaK's selective promiscuity with physics-based modeling." Boston, MA. |

Institutional Seminars

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| 2026 | Invited Seminar , University of Illinois, Springfield, "A Bio-Airlock: Hydration Free-Energy Controls an Ion Channel's Gating," Springfield, IL. |
| 2025 | Invited Seminar , The College of the Holy Cross, "A Bio-Airlock: Hydration Free-Energy Controls an Ion Channel's Gating," Worcester, MA. |

- Invited Seminar**, Chemistry Department Seminar, George Washington University, “Computational design of PROTACs”, Washington D.C.
- 2022 **Invited Seminar**, The College of Idaho Natural Science Symposium, “Computer simulations of proteins help understand their function.” Caldwell, ID.

Awards and Honors

- 2026 **Graduate of the Last Decade (GOLD) Award**, The College of Idaho. For service to the College and outstanding professional achievements.
- 2025 **Excellence in Service Award**, University of Maryland, Baltimore. For outreach in the local Baltimore community and service to the University.
- 2024 **Best of Biophysical Journal**, for publication on BK channel hydrophobic dewetting, titled “Inner pore hydration free energy...”
Outstanding Talk Award, Research Symposium, UM Greenebaum Comprehensive Cancer Center.
- 2022 **Paul H. Terry Endowment Award**, Chemistry Dept., University of Massachusetts Amherst.
- 2020 **William E. McEwen Poster Award**, Chemistry Dept., University of Massachusetts Amherst.
- 2018–19 **Honorable Mention, Top-Gun TA Award**, Chemistry Dept., University of Massachusetts Amherst.
- 2014–18 **Gipson Honors Fellowship**, The College of Idaho.
- 2014 **Kathryn Albertson Scholarship**, Chemistry Dept., The College of Idaho.

Service

- 2026 **Moderator**, Career Panel for UMB Postdoctoral Retreat.
- 2025 **Moderator**, Skills for Faculty Positions panel, University of Maryland Office of Postdoctoral Scholars.
Moderator, Career Panel Discussion, University of Maryland, Baltimore Cancer Retreat.
Poster judge, Frontiers at the Chemistry-Biology Interface Symposium, Johns Hopkins University.
- 2023–24 **Facilitator**, Responsible Conduct of Research NIH training, University of Maryland, Baltimore.
- 2024– **Organization committee, Annual Cancer Research Retreat**, University of Maryland, Baltimore (2x).
- 2020–20 **Organization committee, Alumni Networking Symposium**, Chemistry-Biology Interface program, University of Massachusetts Amherst (2x).
- 2020 **Search committee**, Grad. Program Manager, Chemistry Dept. University of Massachusetts Amherst.
- 2019–21 **ResearchFest committee** for Chemistry Dept., University of Massachusetts Amherst (3x).

Journals Refereed for:

Biophys. J., J. Phys. Chem., ACS Omega, J. Chem. Theory Comput., J. Chem. Inf. Model., Comm. Chem., Drug Discov. Today, ACS Chemical Biology, PROTEINS, PLOS Comput. Biol.

Outreach

- 2023– **Volunteer instructor**, CURE Program, University of Maryland Baltimore, STEM outreach for middle- and high-schoolers in Baltimore ([link](#)) (3x)
Volunteer instructor, Baltimore Underground Science Space (BUGSS) ([link](#)), STEM outreach in Baltimore community-accessible lab space. (1x)
Volunteer instructor, RAMP Program, University of Maryland, Baltimore, STEM outreach for high-schoolers in Baltimore ([link](#)) (2x)
Poster judge, Mount Royal Middle School Science Fair, Baltimore (3x)
Volunteer, Halloween STEM outreach at Community Engagement Center, University of Maryland, Baltimore (2x)
- 2020–23 **Reviewer for Journal of Emerging Investigators**, 25 articles by middle- / high-school students ([link](#))

2022 **Lab for girls summer science camp**, Eureka! at University of Massachusetts Amherst ([link](#))

Professional Development

Society Memberships

2025– American Chemical Society

2022– Biophysical Society

Training and Workshops

2024 **Writing an Effective Teaching Philosophy Statement**, online CIRTl workshop

Teaching Biophysics at a PUI workshop, Biophysical Society Annual Meeting

2024 **Safety preparedness trainings**, CPR/AED, Stop the Bleed, Civilian Active Shooter Events, Fire Extinguishers, and Naloxone Training, University of Maryland, Baltimore ([certificates](#))

2022 **CIRTl associate certification**, University of Massachusetts Amherst, achieved through training on evidence-based and inclusive teaching practices ([link](#))

2021 **Evidence-based Undergraduate STEM Teaching**, online course ([link](#))

Inclusive STEM Teaching, online course ([link](#))

Intellectual Property & Software

- ph4gen: An automated pharmacophore generator for SILCS-Pharm. Copyright disclosure, University of Maryland, Baltimore. 2026. For use with a method licensed to SilcsBio LLC.
- SILCS-xTAC: A method for computationally predicting PROTAC binding. Copyright disclosure, University of Maryland, Baltimore, 2026. For use with a method licensed to SilcsBio LLC.
- SILCS-Hotspots: A method for ranking druggable protein-ligand binding sites. Copyright disclosure, University of Maryland, Baltimore, 2024. For use with a method licensed to SilcsBio LLC.